

LAB 2 — AGILE BACKLOG CREATION & SPRINT SIMULATION

NAME : Priyanka B M

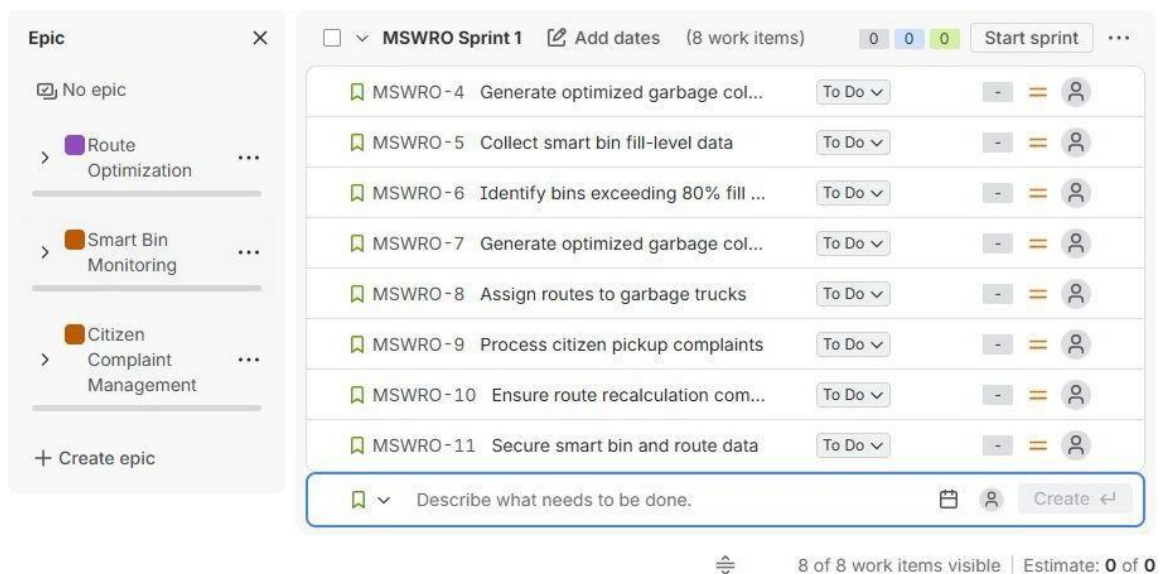
SRN : PES1UG24AM207

DATE : 31/08/2026

SEC : 5D

Problem Statement: Municipal Solid Waste Route Optimizer

1. Epics and User Stories



Epic ID	Epic	User Story ID	User Story
E1	Route Optimization	US1	As a sanitation supervisor, I want the system to generate optimized waste collection routes so that collection vehicles can travel efficiently.
E1	Route Optimization	US2	As a truck driver, I want to view my assigned collection route so that I know

			which bins to collect.
E1	Route Optimization	US3	As a sanitation supervisor, I want routes to be updated when bin conditions change so that vehicles follow efficient routes.
E2	Smart Bin Monitoring	US4	As a sanitation supervisor, I want to view the fill level of bins so that I can identify bins that need collection.
E2	Smart Bin Monitoring	US5	As a sanitation supervisor, I want overflowing bins to be identified automatically so that they can be prioritized for collection.
E2	Smart Bin Monitoring	US6	As a sanitation supervisor, I want to receive alerts for bins reaching critical capacity so that timely action can be taken.
E3	Citizen Complaint Management	US7	As a citizen, I want to report a missed waste collection so that the issue can be addressed.
E3	Citizen Complaint		As a sanitation

	Management	US8	supervisor, I want to view citizen complaints so that unresolved collection issues can be tracked.
E3	Citizen Complaint Management	US9	As a sanitation supervisor, I want to view completed and pending collections so that I can monitor daily waste collection activities.

2. **Sprint board**
Evidence for Sprint 1.

Spaces

Municipal Solid Waste Route Optimizer

Summary Timeline **Board** Backlog Calendar List Forms Development Docs

Complete sprint

To Do 6

Identify bins exceeding 80% fill threshold

Parent
Smart Bin Monitoring

MSWRO-6

Generate optimized garbage collection routes

Parent
Route Optimization

MSWRO-7

Assign routes to garbage trucks

Parent
Route Optimization

MSWRO-8

Process citizen pickup complaints

Parent
Citizen Complaint Management

MSWRO-9

+ Create

In Progress 1

Collect smart bin fill-level data

Parent
Smart Bin Monitoring

MSWRO-5

+ Create

Done 1

Generate optimized garbage collection routes

Parent
Route Optimization

MSWRO-4

+ Create

3. **Sprint 2 . with its duration set to 1 week.**

Start sprint



Required fields are marked with an asterisk *

Sprint name *

MSWRO Sprint 2

Start date *

8/31/2026



3:59 PM



End date *

9/7/2026



3:59 PM



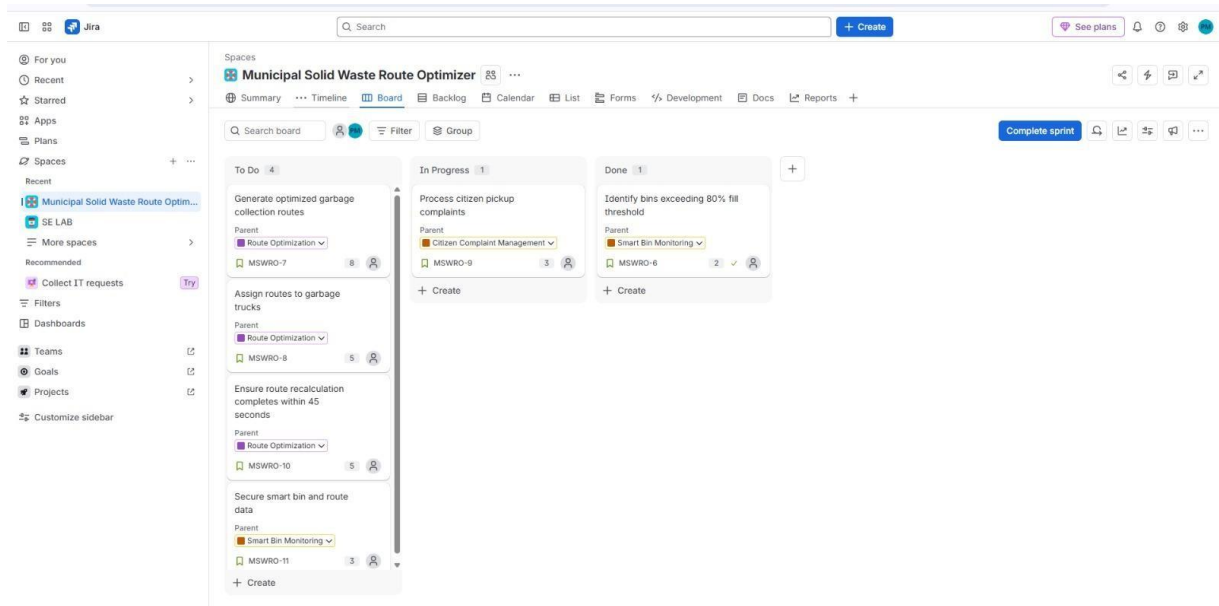
Automatically complete sprint

Sprint goal

Cancel

Start

Evidence for Sprint 2.



4. Sprint 1 Burndown Chart

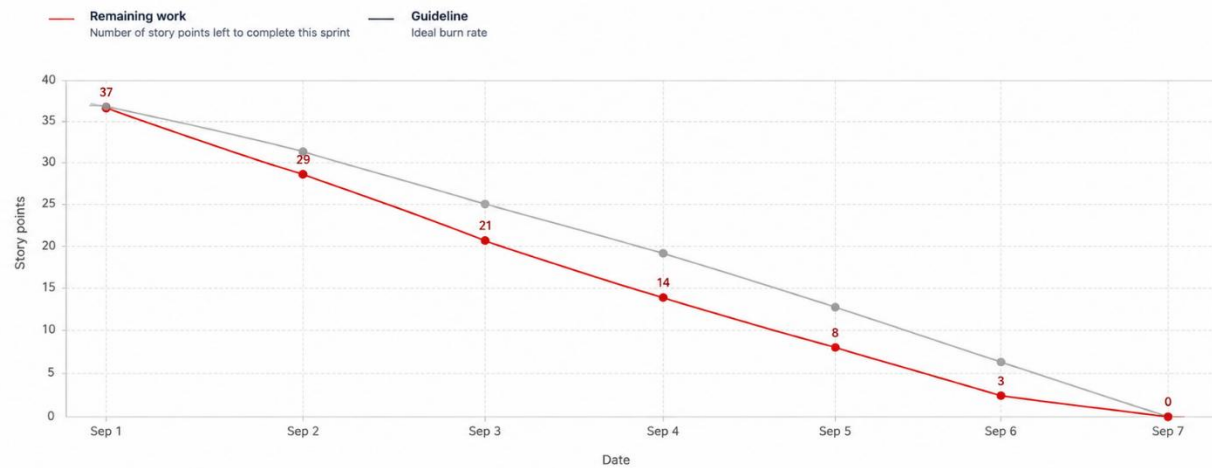
Projects / Municipal Solid Waste Route Optimizer / Reports

Sprint burndown chart

> How to read this report

Sprint: MSWRO Sprint 1
Estimation field: Story points

Date - August 31st, 2026 - September 7th, 2026



Report: MSWRO Sprint 1

*Work item added after sprint start

Scope changes log

[View in All work navigator](#)

Date ↑	Key	Summary	Work type	Epic	Details of scope change	Change in estimation
2026-08-31	MSWRO-4	Generate optimized garbage collectio...	Story	Route O...	Estimate of 8 has been added	- → 8
2026-08-31	MSWRO-5	Collect smart bin fill-level data	Story	Smart B...	Estimate of 3 has been added	- → 3
2026-08-31	MSWRO-6	Identify bins exceeding 80% fill thresh...	Story	Smart B...	Estimate of 2 has been added	- → 2
2026-08-31	MSWRO-7	Generate optimized garbage collectio...	Story	Route O...	Estimate of 8 has been added	- → 8
2026-08-31	MSWRO-8	Assign routes to garbage trucks	Story	Route O...	Estimate of 5 has been added	- → 5
2026-08-31	MSWRO-9	Process citizen pickup complaints	Story	Citizen ...	Estimate of 3 has been added	- → 3
2026-08-31	MSWRO-10	Ensure route recalculation completes ...	Story	Route O...	Estimate of 5 has been added	- → 5
2026-08-31	MSWRO-11	Secure smart bin and route data	Story	Smart B...	Estimate of 3 has been added	- → 3

Your sprint commitment has increased by 37 story points
Due to scope changes: You have 37 story points to complete this sprint

Completed work items

[View in All work navigator](#)

Key	Summary	Work type	Epic	Status	Assignee	Story points
MSWRO-4	Generate optimized garbage collection routes	Story	Route Optimization	Done		8
MSWRO-5	Collect smart bin fill-level data	Story	Smart Bin Monitoring	Done		3

5. Sprint 2 Burndown Chart

Spaces / Municipal Solid Waste Route Optimizer / Reports

Sprint burndown chart

[How to read this report](#)

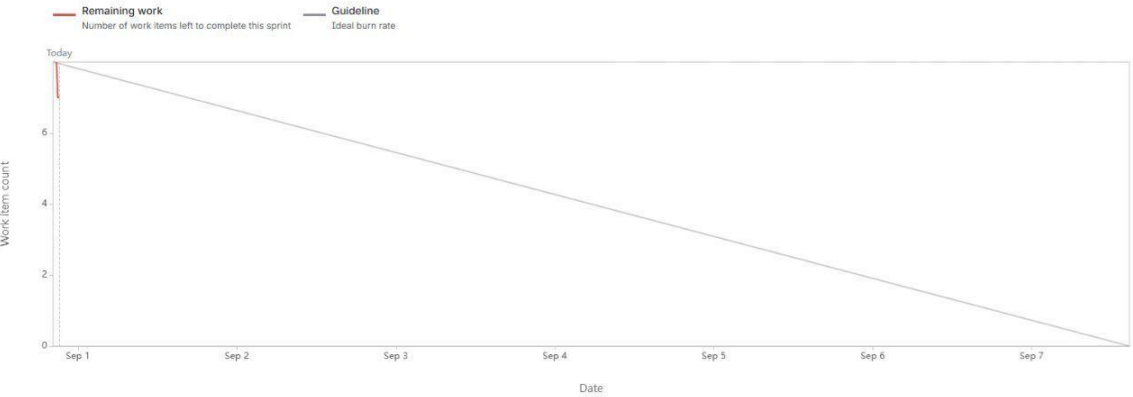
Sprint

MSWRO Sprint 1

Estimation field

Work item count

Date - August 31st, 2026 - September 7th, 2026



Report: MSWRO Sprint 1

*Work item added after sprint start

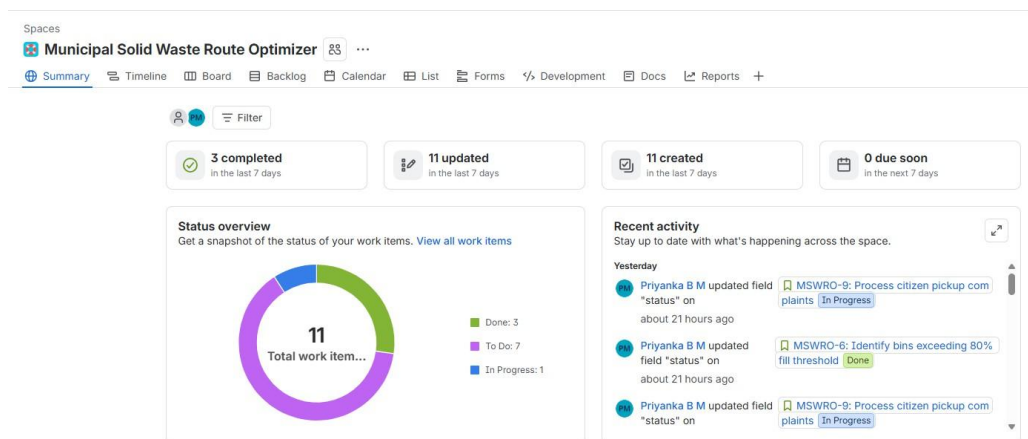
Scope changes log

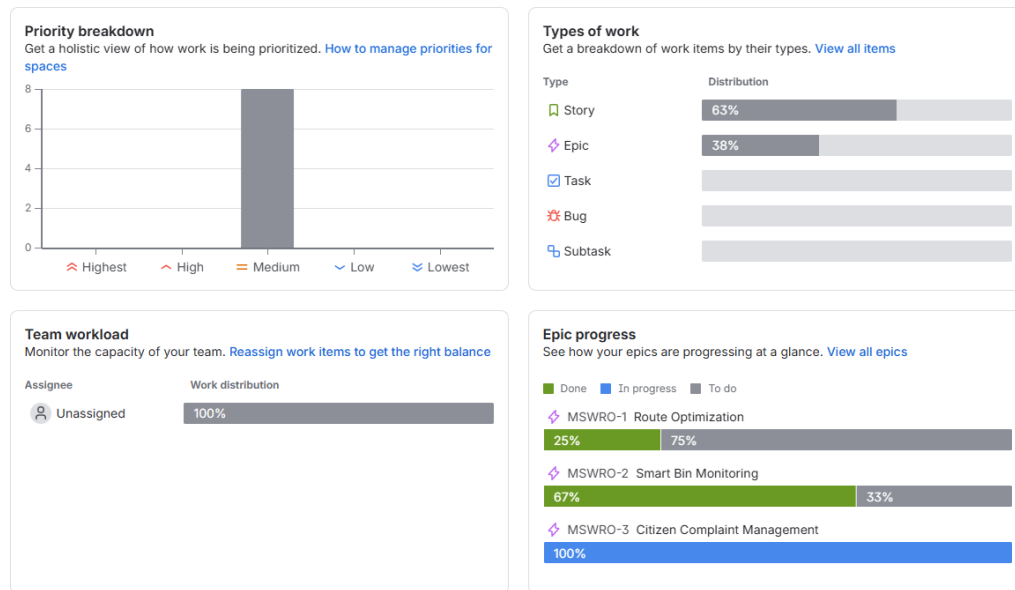
[View in All work navigator](#)

Completed work items							View in All work navigator
Key	Summary	Work type	Epic	Status	Assignee	Story points	
MSWRO-6	Identify bins exceeding 80% fill threshold	Story	Smart Bin Monitoring	Done		2	

Summary:

The Municipal Solid Waste Route Optimizer was organized in Jira using three epics and nine user stories. The user stories were assigned to the appropriate epics, priorities, and story points and organized into one one-week sprint. The sprint board was used to demonstrate the progress of tasks through To Do, In Progress, and Done. The Burndown Chart was used to track the remaining work and progress throughout the sprint. This Jira-based sprint simulation helped demonstrate Agile backlog management, sprint planning, task tracking, and progress monitoring.





Group Reflection.

1. How well did the team work together during the sprint?

Our team worked together effectively by dividing the work, discussing the user stories, and tracking the progress of the waste management system in Jira.

2. Was Planning Poker useful for estimating the user stories?

Yes. Planning Poker helped us estimate the complexity of features such as **route optimization, smart bin monitoring, and citizen complaint management** and reach a common decision as a team.

3. How did the team prioritize the user stories?

We prioritized the stories based on their importance to the Municipal Solid Waste Route Optimizer. Features related to **monitoring bin levels and optimizing collection**

routes were given higher priority because they are essential to the main functionality of the system.

4. What challenges did the team face during the sprint?

The main challenges were estimating the complexity of some stories and managing the tasks within the sprint duration. We resolved these through discussion and by dividing the work among team members.

5. How did Jira help in completing the sprint?

Jira helped us organize the three epics, manage the user stories, assign story points, and track their progress from **To Do → In Progress → Done**.

6. What did the Burndown Chart help the team understand?

The Burndown Chart helped us monitor how quickly the selected waste-management user stories were being completed and compare our actual progress with the planned sprint progress.

7. What could the team improve in the next sprint?

We could improve our **time management, estimation accuracy, and task distribution**. We could also communicate more frequently while working on route optimization, smart-bin monitoring, and complaint management features.

8. What did the team learn from this sprint?

We learned how Agile practices can be applied to a real-world **municipal waste management system**. The activity helped us understand backlog prioritization, story-point estimation, sprint planning, Jira tracking, and monitoring sprint progress using a Burndown Chart.

During the sprint, our team worked together to plan, prioritize, and complete the user stories for the **Municipal Solid Waste Route Optimizer**. We used **Planning Poker** to estimate the complexity of features related to **Route Optimization, Smart Bin Monitoring, and Citizen Complaint Management** and discussed the estimates as a team. We prioritized the backlog based on the importance of each feature and selected suitable stories for Sprint 1. Jira helped us track the progress of the stories from **To Do → In Progress → Done** and monitor our work using the **Burndown Chart**. Overall, this sprint helped us understand how Agile practices can be applied to a real-world waste management system. We also learned that better communication, time management, task distribution, and accurate estimation could improve our performance in future sprints.